

Opgave 1 – Perslucht

a)

Bijvoorbeeld: - Grotere krachten, beweging beter te controleren.

2P

b)

Doordat olie niet goed samendrukbaar is.

①

1P

c)

$$10 \text{ m}^3 \cdot 0,75\% = \pm 18 \text{ g} = 135 \text{ g} \text{ (20}^\circ\text{C)} \text{ ①}$$

$$10 \text{ m}^3 \cdot 100\% = \pm 90 \text{ g} = 90 \text{ g} \text{ (10}^\circ\text{C)} \text{ ①}$$

$$135 - 90 = 45 \text{ g} \text{ ①}$$

4P

d)

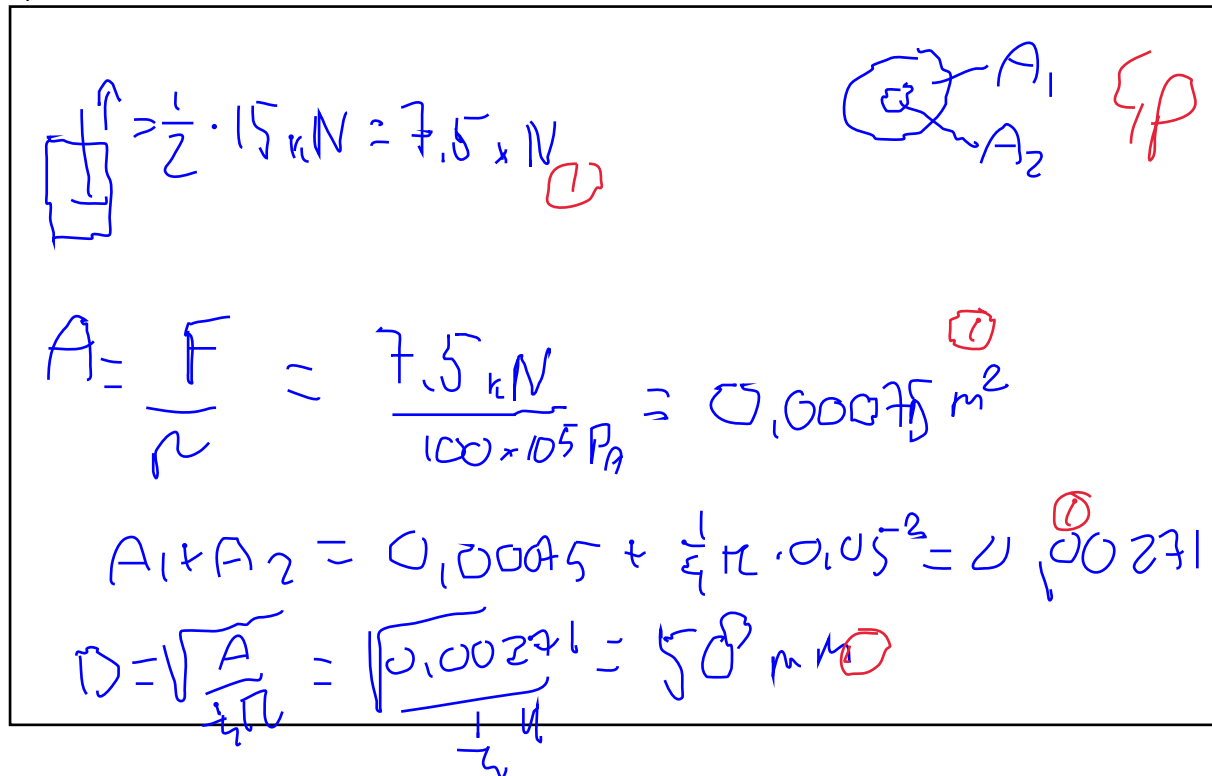
$$P_1 \cdot V_1 = P_2 \cdot V_2 \Rightarrow \text{DRUK} = \underline{1 \text{ BAR}} \text{ ①}$$

$$F = p \cdot A$$

$$F = 1 \times 10^5 \cdot \frac{1}{4} \pi \cdot 0,2^2 = 3142 \text{ N} \text{ ① ①}$$

3P

e)

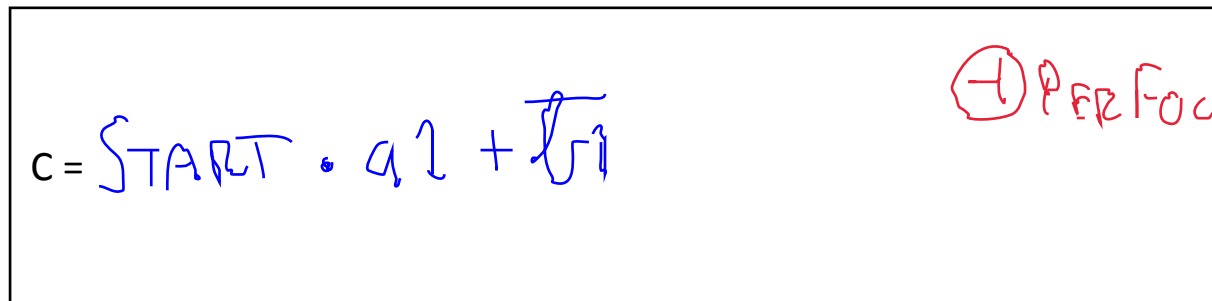


Handwritten calculations and diagrams for a pneumatic cylinder problem:

- Diagram of a cylinder with force F and area A indicated.
- Equation: $F = \frac{1}{2} \cdot 15 \text{ kN} = 7.5 \text{ kN}$ (circled in red)
- Diagram of a cylinder with areas A_1 and A_2 indicated.
- Equation: $A = \frac{F}{p} = \frac{7.5 \text{ kN}}{100 \times 10^5 \text{ Pa}} = 0.00075 \text{ m}^2$ (circled in red)
- Equation: $A_1 + A_2 = 0.00075 + \frac{1}{4} \pi \cdot 0.05^2 = 0.00271$ (circled in red)
- Equation: $D = \sqrt{\frac{A}{\frac{1}{4} \pi}} = \sqrt{\frac{0.00271}{\frac{1}{4} \pi}} = 58 \text{ mm}$ (circled in red)

Opgave 2 - Pneumatisch schakelen

a)



Handwritten formula for a pneumatic cylinder:

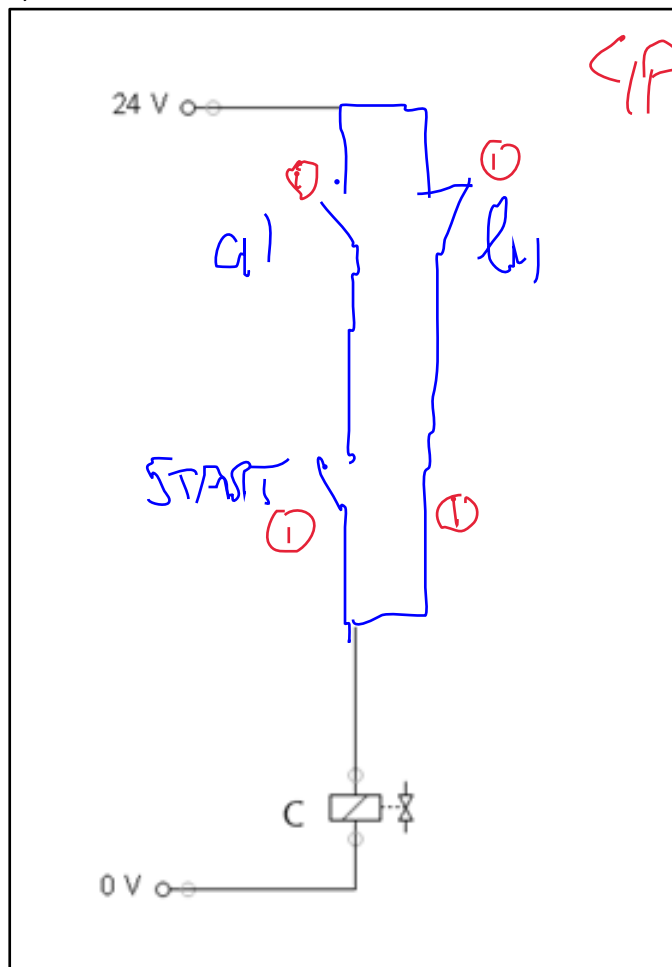
$$C = S_{\text{START}} \cdot a_1 + \sqrt{1}$$

(circled in red)

Naam:

Studentnummer:

b)



c)

b1	start	a1	C
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	⊖
1	0	1	⊖
1	1	0	⊖
1	1	1	

2P

Opgave 3 – Volgordebesturing

a)

2P

A+, B+ ,C+ ,C- ,A- , B-

-1 P/FOUT

b)

3P

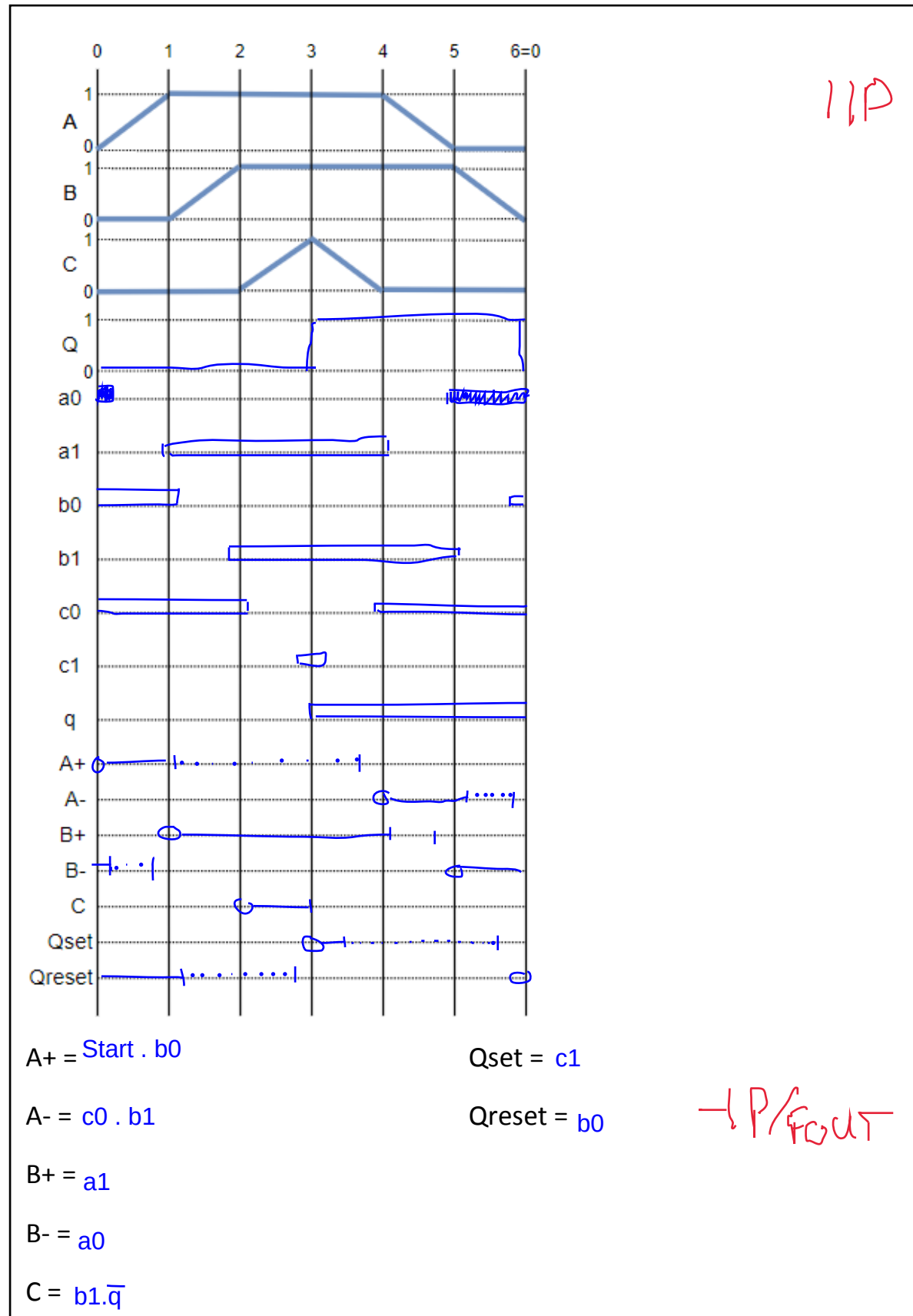
Step	0	1	2	3	4	5	6=0
A	0	1	1	1	1	0	0
B	0	0	1	1	1	1	0
C	0	0	0	1	0	0	0

0	1	3	7	3	2	0
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SET OP 3 ①

RESET OP 6=0 ①

c)



Opgave 4 – Stromingsleer

a)

$$A_1 \cdot v_1 = A_2 \cdot v_2 \quad (1)$$

$$q = A \cdot v \quad (1)$$

5P

$$v = \frac{q}{A} = \frac{175/60/1000}{\frac{1}{4} \pi 0,04^2} = 2.321 \text{ m/s} \quad (1)$$

$$v_2 = \frac{A_1 \cdot v_1}{A_2} = \frac{\frac{1}{4} \pi 0,04^2 \cdot 2.321}{\frac{1}{4} \pi 0,007^2} \quad (1)$$

1182 m/s (1)

b)

$$Re = \frac{\rho \cdot v \cdot d}{\eta} \Rightarrow v = \frac{Re}{\frac{\rho \cdot d}{\eta}} \quad (1)$$

4P

$$v = \frac{3500}{\frac{1000 \cdot 0,001}{0,001}} = 0,7 \text{ m/s} \quad (1)$$

c)

$$P_v = f \cdot \frac{l}{d} \cdot \frac{1}{2} \cdot \rho \cdot v^2 \quad (1)$$

4P

$$l = 20 + 2 \cdot (40 \cdot 0.005) = 20.4 \quad (?)$$

$$P_w = 0.025 \cdot \frac{20.4}{0.005} \cdot \frac{1}{2} \cdot 1000 \cdot 1^2 = 5100 \text{ Pa} \quad (1)$$

51 kPa

0.51 bar

Vrije ruimte: